Victor J. Huarcaya Azañon

Bern, Switzerland (B permit) | vhuarcaya.github.io | victor.huarcaya@proton.me | linkedin.com/in/vhuarcaya

Summary

Optical engineer currently leading optical AIV for ESA's Comet Interceptor instrument Comet Camera (CoCa), with 10+ years of experience in precision optics, interferometry, and optomechanical systems. Background includes high-stability interferometric testbeds and engineering model development supporting ESA's *LISA* gravitational wave observatory. Skilled in tolerance-driven optical integration, cleanroom workflows (ISO 5/7), and ECSS-compliant verification planning. Strong at the optics–mechanics–electronics interface, with proven ability to translate scientific requirements into traceable, measurable acceptance criteria.

Professional Experience

Optical AIV Engineer — Sept. 2024 – Present
Bern University — Bern, Switzerland

- Lead optical AIV activities for the Comet Camera (CoCa) on ESA's F-class *Comet Interceptor* mission through Phases C/D/E; established and commissioned ISO 5 cleanroom integration facility.
- Delivered flight-representative optical bench on schedule for CDR-level environmental testing, meeting all optical performance requirements (WFE, MTF, co-alignment).
- Authored and released AIT procedures, test scripts, and verification reports under ECSS configuration control; manage non-conformance reports (NCRs) and coordinate rework to closure.
- Drive technical documentation across multi-site AIV teams, ensuring traceability from instrument requirements to acceptance criteria.

Postdoctoral Researcher — Oct. 2023 – Sept. 2024
Max Planck Institute for Gravitational Physics (Albert Einstein Institute) — Hannover, Germany

- *LISA* (Laser Interferometer Space Antenna) Phase Measurement Subsystem and *LISA* engineering model: optical payload Assembly, Integration and Testing.
- Successfully integrated and tested *LISA* Phase Measurement Subsystem, delivering key project milestones for ESA's space-based gravitational wave observatory.

Research Scientist — May 2017 – Oct. 2023
Max Planck Institute for Gravitational Physics (Albert Einstein Institute) — Hannover, Germany

- Achieved world-record sub-picometer structural stability ($< 1\,\mathrm{pm}/\sqrt{\mathrm{Hz}}$ below 1 mHz) in compact interferometers, advancing precision for future space missions.
- Developed 5-DOF test mass optical readout system delivering $10\times$ higher angular precision than commercial autocollimators at low frequencies.
- Designed and validated novel single-element dual-interferometer architecture, reducing system complexity while maintaining sub-picometer performance.
- Operated and maintained ultra-high vacuum systems (10^{-7} mbar) for precision interferometry experiments.

Spacecraft Controller — March 2016 – May 2017
EUMETSAT (for SERCO) — Darmstadt, Germany

- Monitored and commanded *Meteosat-7* spacecraft in 24/7 operational environment, maintaining mission performance through end-of-life phase.
- Executed spacecraft decommissioning procedures as part of mission close-out team.

Research Assistant — Sept. 2013 – Jan. 2016
Centre for Quantum Technologies — Singapore

- Designed, aligned, and characterized electro-optical and optomechanical systems for quantum optics experiments.
- Implemented FPGA-based PID controllers for laser frequency stabilization, improving lock stability and bandwidth.

Education and Certifications

Ph.D. in Physics (Dr. rer. nat.) 2023
Leibniz Universität Hannover Hannover, Germany

- **Thesis:** *Advancements in Optical Readout Technologies: Test Mass Sensing and Laser-Frequency Stabilization Techniques for Optical Compact Interferometry*
- Research conducted at Max Planck Institute for Gravitational Physics (Albert Einstein Institute)

MSc. in Space Science and Technology 2013
Luleå University of Technology Luleå, Sweden

- **Thesis:** *Inertial Electrostatic Confinement Devices as an Advanced Electric Propulsion System*
- Research conducted at Institute of Space Systems (IRS) - Stuttgart

MSc. in Space Techniques and Instrumentation 2012
Université Paul Sabatier – Toulouse III Toulouse, France

***Licenciatura* in Physics (240 ECTS credits, equivalent to BSc. + MSc.)** 2010
University of Balearic Islands Palma, Spain

ESA's International School on Space Optics – Certification (ESTEC, Noordwijk), 2025
ESA's Space Optics Instrument Design and Technology Course – Certification (Poltu Quatu, Italy), 2023
Laser Safety Officer Certification – LZH Laser Akademie (Hannover, Germany), 2023

Skills

Languages: Spanish (Native) | English (C1 – Advanced) | German (A1 – Basic)

Technical Skills:

- **Opto-Mechanical Integration:**
  - End-to-end AIT of optical payloads: telescope structures, interferometric benches, focal plane assemblies
  - Precision positioning systems: hexapods, piezo stages, goniometers, kinematic mounts
  - Environmental test support: TVAC, thermal cycling, vibration campaigns with pre/post optical verification
  - Tolerance budgeting: flow-down from instrument WFE/MTF requirements to component-level alignment specs
- **Optical Metrology:**
  - Alignment tools: theodolites, autocollimators, shear-plate interferometers, alignment telescopes
  - Wavefront sensing: Shack–Hartmann sensors, Fizeau/Twyman-Green interferometers (Zygo-type), PSF/MTF analysis
  - 3D coordinate metrology: `Spatial Analyzer` software for theodolite-based measurements
- **Electro-Optics & Detection:**
  - Photodetector systems: single-element PDs, QPDs, low-noise transimpedance amplifiers
  - Laser systems: frequency stabilization, AOM/EOM modulation, Class 3B/4 safety compliance
  - Signal chain: ADC/DAC integration, noise characterization (shot noise, electronic noise floors)
- **Software & Analysis:**
  - Optical design: `Zemax OpticStudio` (ray tracing, tolerance analysis, optimization)
  - Data analysis & automation: `MATLAB`, `Python` (NumPy, SciPy, instrument control APIs)
  - Documentation: `LaTeX`, `MS Office`, `Adobe Illustrator` for technical schematics
  - Lab control: Serial/GPIB/TCP-IP instrument communication, digital/analog I/O
- **Project Management:**
  - Preparation of AIT plans, test procedures, alignment reports, and configuration logs
  - Experience with ECSS / ESA project workflows
  - Procurement and technical coordination with optical and lab-equipment suppliers

Publications

1. **V. Huarcaya** et al. "2×10^{-13} fractional laser-frequency stability with a 7-cm unequal-arm Mach-Zehnder interferometer". *Physical Review Applied*, 20, 024078, 2023. doi:10.1103/PhysRevApplied.20.024078
2. **V. Huarcaya** et al. "Single-Element Dual-Interferometer for Precision Inertial Sensing: Sub-Picometer Structural Stability and Performance as a Reference for Laser Frequency Stabilization". *Sensors*, 23(24):9758, 2023. doi:10.3390/s23249758
3. **V. Huarcaya** et al. "Five degrees of freedom test mass readout via optical levers". *Classical and Quantum Gravity*, 37(2):025004, 2019. doi:10.1088/1361-6382/ab5c73
4. Y. Yang, K. Yamamoto, **V. Huarcaya** et al. "Single Element Dual Interferometer for precision inertial sensing". *Sensors*, 20(17):4986, 2020. doi:10.3390/s20174986
5. S. Aria, [...], **V. Huarcaya** et al. "GLINT – Gravitational-wave laser INterferometry triangle". *Experimental Astronomy*, 44, 181–208, 2017. doi:10.1007/s10686-017-9558-x
6. **V. Huarcaya**. "Advancements in optical readout technologies: Test mass sensing and laser-frequency stabilization techniques for optical compact interferometry". *PhD Thesis*, Leibniz Universität Hannover, 2024. doi:10.15488/16938

Selected Presentations

International Conference on Space Optics - ICSO – "Laser-frequency Stabilization via Locking to a Single-Element Mach-Zehnder Interferometer" (Antibes Juan-les-Pins, France), 2024

International Conference on Space Optics - ICSO – "2e-13 fractional laser frequency stability with a 7 cm unequal-arm Mach-Zehnder interferometer" (Antibes Juan-les-Pins, France), 2024

13th LISA Symposium – "Five degrees of freedom Test Mass readout via optical levers" (Online), 2020

12th LISA Symposium – "Testing an Optical Readout scheme for future Gravitational Reference Sensors" (Chicago), 2018

Invited Talk – "Demonstration of multi-degree of freedom Test Mass readout" (Wuhan, China), 2018

Alpbach Summer School – "Quantum Physics and Fundamental Physics in Space" (ESA/FFG, Austria), 2015

Scholarships

Graduate Research Scholarship – National University of Singapore, 2014

Erasmus Mundus Mobility Grant – European Commission, 2012

Sa Nostra Foundation Grant – For MSc. studies abroad, Spain, 2010–2011

SENECA Scholarship – Spanish Ministry of Education, 2008

Early Career Highlight

ESA Student Parabolic Flight Campaign – Flight Team Leader for experiment "*Visualization of the Richtmyer-Meshkov instability in zero-g*" (Bordeaux, France), 2004